The empirical recurrence for OEIS A207276, proved by transfer matrix

Adrian Perez Fontelles
Independent researcher

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Abstract

OEIS A207276 counts arrays of width 1 over $\{0, 1, 2, 3\}$ containing no occurrence of certain short patterns, read along the rows, the columns or the diagonals. The entry carries an empirical recurrence of order 5 contributed by R. H. Hardin. It is true, and it is decidable rather than empirical: no pattern is longer than 3 letters and no direction moves the row index by more than one, so every occurrence lies inside 3 consecutive array rows and the count is a walk count on $S = 21$ states.

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1 The conjecture

OEIS A207276 is "Number of n X 1 0..3 arrays avoiding the patterns z z+1 z or z z-1 z in any row, column, diagonal or antidiagonal.". It has offset 1 and begins

$$4, 16, 58, 214, 788, 2902, 10686, 39350, \ \ldots$$

The entry states, as a conjecture contributed by its author and never marked settled:

Empirical: a(n) = 3*a(n-1) + 2*a(n-2) + a(n-3) + 3*a(n-4) + a(n-5).

As of the "Last modified" line on the live entry (Feb 20 2018 14:26:30, revision 8) this is still recorded as empirical, and nothing on the entry records it as proved.

2 What is being counted

The array has 1 cells across, with entries in $\{0, 1, 2, 3\}$. An OCCURRENCE of a pattern in a direction is a run of consecutive cells taken in that direction whose entries match the pattern in order; the array is counted when it has no occurrence at all. The entry names

direction	forbidden patterns
along a row, left to right	$(z,\, z+1,\, z)$, $(z,\, z-1,\, z)$
down a column	$(z,\, z+1,\, z)$, $(z,\, z-1,\, z)$
down a diagonal, nw to se	$(z,\, z+1,\, z)$, $(z,\, z-1,\, z)$
down an antidiagonal, ne to sw	$(z,\, z+1,\, z)$, $(z,\, z-1,\, z)$

A pattern written with the free letter z forbids every window standing in those differences, whatever the value of z: the window $(v_1, \ldots, v_r)$ is an occurrence exactly when the numbers $v_s - o_s$ are all equal, o being the offsets shown.

3 Every occurrence lies in 3 consecutive rows

Lemma 1. *Each of the directions above changes the row index by* 0 *or* 1 *from one cell of a run to the next, and no pattern is longer than* 3 *letters. Hence every occurrence lies inside* 3 *consecutive array rows, and an occurrence whose last cell is in row i lies inside rows $i-2,\dots,i$.*

Proof. Immediate from the offsets listed: the row step is 0 for a row and 1 for the other three. □

So the state is the window of the last 2 array rows, a row still to come being written $\perp$. A step appends a row and rejects it exactly when some occurrence ends in it or lies inside it; an occurrence needing a row that is $\perp$ cannot exist, which is the boundary rule and is not assumed but read off the window. The walk starts at the all-$\perp$ window, so a walk of no steps is the empty array and a walk of R steps is an array of R rows. With M the adjacency matrix, ι the indicator of the starting window and $\tau = \mathbf{1}$,

$$a(n) \;=\; \iota^{\top} M^{\,j} \tau, \qquad S = 21.$$

The walk index j and the entry's own n differ by a constant, read off by matching the published terms rather than assumed. In particular a is C-finite.

4 The proof

Lemma 2. *Let $q(t) = t^5 - \sum_i c_i t^{5-i}$ be the characteristic polynomial of the conjectured recurrence. The residual $a(n) - \sum_i c_i a(n-i)$ equals $\iota^{\top} M^{\,j} q(M)\tau$ for the corresponding walk index j, so the recurrence holds from some point on if and only if $\iota^{\top} M^{j} q(M)\tau = 0$ for all large j, and it is enough to examine $j < S$.*

Proof. Factor M^j out of $M^{j+5} - \sum_i c_i M^{j+5-i}$. For the bound, M^S is an integer combination of $I, M, \dots, M^{S-1}$ by Cayley–Hamilton, so the numbers $u_j = \iota^{\top} M^j q(M)\tau$ obey the monic recurrence given by the characteristic polynomial of M; S consecutive zeros force every later one, and the last nonzero u_j pins the threshold exactly. □

Theorem 3.

$$a(n) \;=\; 3\,a(n-1) + 2\,a(n-2) + a(n-3) + 3\,a(n-4) + a(n-5)$$

for every $n > 5$.

Proof. The vector $w = q(M)\tau$ was formed by 5 matrix–vector products in exact integer arithmetic and $\iota^{\top} M^j w$ evaluated for $j = 0, 1, \dots,$ again exactly; those integers vanish from the index corresponding to $n = 5 + 1$ onwards and the run continued until the working vector was identically zero, which settles every larger j at once. □

5 Verification

Four checks, each independent of the algebra above.

First, the model reproduces all 24 terms the entry publishes, exactly, in integer arithmetic.

Second, the count was recomputed for the smallest arrays straight from the definition: fill the cells one at a time, in the orientation the entry's own name writes, and reject as soon as a completed run matches a forbidden pattern. No transfer matrix, no window and no transposition are involved, and the published terms came back.

Third, the conjectured recurrence was evaluated directly on the published terms, with no matrices involved, and holds wherever the proved range and the data overlap.

Fourth, the annihilation test of Lemma 2 was carried out in exact integer arithmetic on the whole vector rather than on a sampled prefix.

References

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[4] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.